

Derivation of the Tripartite S_{21} Transmission Spectrum

1 Equations of motion

We work in a frame rotating at the probe frequency ω , so each mode is described by how far it sits from the probe:

$$\Delta_j = \omega_j - \omega, \quad j \in \{c, m, q\}. \quad (1)$$

ω frequency of the microwave tone sent in by the VNA

$\omega_c, \omega_m, \omega_q$ bare frequencies of the cavity, magnon and qubit

$\Delta_c, \Delta_m, \Delta_q$ detunings; a mode is on resonance with the probe when its Δ is zero

The three-mode Hamiltonian in this frame is

$$\frac{\hat{H}_{\text{sys}}}{\hbar} = \Delta_c c^\dagger c + \Delta_m m^\dagger m + \Delta_q q^\dagger q + g_{cm}(c^\dagger m + cm^\dagger) + g_{cq}(c^\dagger q + cq^\dagger). \quad (2)$$

c, m, q annihilation operators of the cavity photon, the magnon and the qubit excitation

g_{cm} cavity–magnon coupling strength

g_{cq} cavity–qubit coupling strength

Two simplifications are built into Eq. (2). First, the qubit is written as a harmonic mode q , which is allowed because the probe power is very low, so the transmon only moves between $|0\rangle$ and $|1\rangle$ and never feels its own anharmonicity. Second, the direct magnon–qubit coupling g_{mq} has been dropped, because the transmon and the YIG film are on different regions of the cavity and do not talk to each other directly.

Each mode evolves according to the Heisenberg–Langevin equation,

$$\dot{\hat{o}} = -\frac{i}{\hbar} [\hat{o}, \hat{H}_{\text{sys}}] - \frac{\Gamma_o}{2} \hat{o} + \sqrt{\Gamma_{o,\text{ext}}} \hat{o}_{\text{in}}, \quad (3)$$

$\hat{o}$ any one of the three mode operators c, m, q

Γ_o total loss rate of that mode

$\Gamma_{o,\text{ext}}$ part of the loss through which the mode is driven from outside

$\hat{o}_{\text{in}}$ field arriving at the mode from outside

The commutators needed for the cavity are

$$[c, c^\dagger c] = c, \quad [c, c^\dagger m + cm^\dagger] = m, \quad [c, c^\dagger q + cq^\dagger] = q, \quad (4)$$

so that Eq. (3) applied to each mode in turn gives

$$\dot{c} = -\left(i\Delta_c + \frac{\kappa_c}{2}\right)c - ig_{cm}m - ig_{cq}q + \sqrt{\kappa_1}a_{\text{in}}, \quad (5)$$

$$\dot{m} = -\left(i\Delta_m + \frac{\kappa_m}{2}\right)m - ig_{cm}c, \quad (6)$$

$$\dot{q} = -\left(i\Delta_q + \frac{\gamma_q}{2}\right)q - ig_{cq}c. \quad (7)$$

κ_c total cavity loss rate

κ_m magnon loss rate, set by the Gilbert damping of the YIG

γ_q qubit relaxation rate

κ_1 rate at which the cavity is coupled to Port 1

a_{in} amplitude of the microwave drive entering at Port 1

The cavity loss rate appearing in Eq. (5) is the total one,

$$\kappa_c = \kappa_1 + \kappa_2 + \kappa_{\text{int}}, \quad (8)$$

since photons leak out of both ports and into the walls, whereas the drive term carries only κ_1 , because we drive from Port 1 alone. The noise inputs of the magnon, the qubit and the undriven port have been left out of Eqs. (5)–(7) because they average to zero, and S_{21} depends only on average values.

2 Steady-state solution

With a continuous-wave drive, the system eventually settles and nothing changes any more, so

$$\dot{c} = \dot{m} = \dot{q} = 0. \quad (9)$$

Applying this to Eqs. (6) and (7) gives the magnon and the qubit directly in terms of the cavity field,

$$m = \frac{-ig_{cm}}{i\Delta_m + \frac{\kappa_m}{2}} c, \quad q = \frac{-ig_{cq}}{i\Delta_q + \frac{\gamma_q}{2}} c. \quad (10)$$

In other words, neither the magnon nor the qubit does anything by itself; both are simply driven by the cavity. On resonance the magnon amplitude becomes

$$m|_{\Delta_m=0} = -\frac{2ig_{cm}}{\kappa_m} c, \quad (11)$$

which is purely imaginary relative to c , i.e. the magnon lags the cavity by 90° . This is what eventually produces the sharp dip in transmission.

Substituting Eq. (10) back into Eq. (5), we get:

$$0 = -(i\Delta_c + \frac{\kappa_c}{2}) c - \frac{g_{cm}^2}{i\Delta_m + \frac{\kappa_m}{2}} c - \frac{g_{cq}^2}{i\Delta_q + \frac{\gamma_q}{2}} c + \sqrt{\kappa_1} a_{\text{in}}. \quad (12)$$

Rearranging the equation gives :

$$c = \frac{\sqrt{\kappa_1} a_{\text{in}}}{i\Delta_c + \frac{\kappa_c}{2} + \frac{g_{cm}^2}{i\Delta_m + \frac{\kappa_m}{2}} + \frac{g_{cq}^2}{i\Delta_q + \frac{\gamma_q}{2}}}. \quad (13)$$

3 Input-output relation and S_{21}

The field leaving a port is the field going in, minus the part that leaks out of the cavity:

$$a_{\text{out},j} = a_{\text{in},j} - \sqrt{\kappa_j} c. \quad (14)$$

In a transmission measurement nothing is fed into Port 2, so we take $a_{\text{in},2} = 0$ and

$$a_{\text{out},2} = -\sqrt{\kappa_2} c. \quad (15)$$

The transmission ratio is given as,

$$S_{21} = \frac{a_{\text{out},2}}{a_{\text{in}}}. \quad (16)$$

Substituting Eqs. (13) and (15) into Eq. (16), the drive amplitude a_{in} cancels out and the result is

$$S_{21} = -\sqrt{\kappa_1 \kappa_2} \left[\left(i\Delta_c + \frac{\kappa_c}{2} + \frac{g_{cm}^2}{i\Delta_m + \frac{\kappa_m}{2}} \right) + \frac{g_{cq}^2}{i\Delta_q + \frac{\gamma_q}{2}} \right]^{-1}. \quad (17)$$

κ_2 is the rate at which the cavity is coupled to Port 2

$\sqrt{\kappa_1 \kappa_2}$ is the overall amplitude factor

Multiplying out the nested fractions gives the same expression in a much neater form given as:

$$S_{21} = \frac{-\sqrt{\kappa_1 \kappa_2} (i\Delta_m + \frac{\kappa_m}{2}) (i\Delta_q + \frac{\gamma_q}{2})}{(i\Delta_c + \frac{\kappa_c}{2}) (i\Delta_m + \frac{\kappa_m}{2}) (i\Delta_q + \frac{\gamma_q}{2}) + g_{cm}^2 (i\Delta_q + \frac{\gamma_q}{2}) + g_{cq}^2 (i\Delta_m + \frac{\kappa_m}{2})}. \quad (18)$$

When written in this way the physics becomes easier to understand. The denominator is a cubic, so it has three roots, and these are the three hybridised peaks seen in the transmission map. The numerator vanishes when $\Delta_m \approx 0$ or $\Delta_q \approx 0$, so the transmission drops to nearly zero whenever the probe hits the bare magnon or qubit frequency. Since these two conditions involve neither g nor Δ_c , the dips stay put at the bare frequencies even when the peaks around them shift, which is why the antiresonance is such a reliable way to read off the qubit frequency. The dips are not infinitely deep only because κ_m and γ_q are non-zero.